

ENGINEERING SPECIFICATIONS

Document Name: KAS Double Circuit Data Centre Modbus List		
Document No: ESP-25-26-01	Revision: R1	Date:06-05-2025
Prepared By: PPB	Checked By: NMR	Approved By: NMR

KAS Double Circuit Data Centre				
Descriptions	UOM	Type	Read/Write	Modbus Add(Holding Register)
Evaporator Water In Temperature	°C	ANALOG	Read	40002
Evaporator Water Out Temperature	°C	ANALOG	Read	40003
LWT Setpoint	°C	ANALOG	Read/Write	40151
Ambient Temperature	°C	ANALOG	Read	40008
Suction Pressure	Bar	ANALOG	Read	40010
Suction Temperature	°C	ANALOG	Read	40006
Suction Saturation Temperature	°C	ANALOG	Read	40015
Suction Superheat	°C	ANALOG	Read	40014
Evaporator Approach	-	ANALOG	Read	40022
Sump Oil Temperature	°C	ANALOG	Read	40118
Liquid Line Temperature	°C	ANALOG	Read	40007
Chilled Water Bypass Valve Status	-	INTEGER	Read	45255
Chilled Water Outlet Valve Status	-	INTEGER	Read	45254
Chilled Water Flow (LPM)	-	ANALOG	Read	40019
Chiller Primary Pump Stratus	-	INTEGER	Read	45256
Adiabatic Setpoint	°C	ANALOG	Read/Write	40049
Adiabatic Status	-	Boolean	Read	10194
Chiller Common Alarm	-	Boolean	Read	10195
Communication Fail	-	Boolean	Read	10196
Chiller Main Switch On Off Status	-	Boolean	Read	10016
Chiller Water Flow Alarm Status	-	Boolean	Read	10014
Outdoor Air Temperature	°C	ANALOG	Read	40008
Fan Speed Status Command	%	INTEGER	Read	45206
EM Kw	-	INTEGER	Read	45382
EM R Current	-	INTEGER	Read	45383
EM Y Current	-	INTEGER	Read	45384
EM B Current	-	INTEGER	Read	45385
EM N Current	-	INTEGER	Read	45386
EM Frequency	-	INTEGER	Read	45387
EM KVA	-	INTEGER	Read	45388
EM KVAR	-	INTEGER	Read	45389
EM Avg Volts	-	INTEGER	Read	45390
Fan Fault	-	Boolean	Read	10190
Alarm Reset	-	Boolean	Read/Write	00111
Compressor - 1				
Discharge Temperature	°C	ANALOG	Read	40005
Discharge Saturation Temperature	°C	ANALOG	Read	40016
Discharge Superheat	-	ANALOG	Read	40025
Discharge Pressure	Bar	ANALOG	Read	40011
Oil Feed Pressure	Bar	ANALOG	Read	40012

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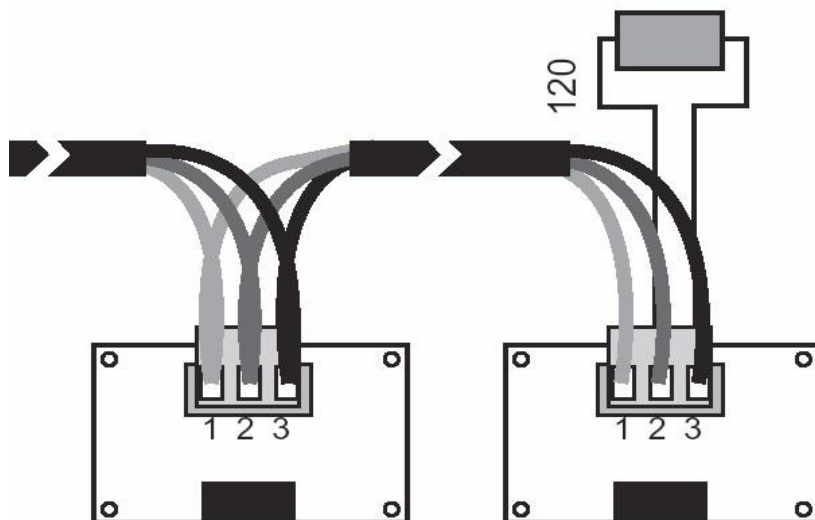
Compressor Current	Amp	ANALOG	Read	40013
Current Percentage Loading	%	ANALOG	Read	40021
Run Hours- High*	Hrs.	INTEGER	Read	45091
Run Hours- Low*	Hrs.	INTEGER	Read	45090
Number Of Starts H*	-	INTEGER	Read	45092
Number Of Starts L*	-	INTEGER	Read	45093
Compressor Status	-	Boolean	Read	10020
Run Hours Total	-	INTEGER	Read	45392
Compressor – 2				
Discharge Temperature	°C	ANALOG	Read	40035
Discharge Saturation Temperature	°C	ANALOG	Read	40046
Discharge Superheat	-	ANALOG	Read	40055
Discharge Pressure	Bar	ANALOG	Read	40041
Oil Feed Pressure	Bar	ANALOG	Read	40042
Compressor Current	Amp	ANALOG	Read	40043
Current Percentage Loading	%	ANALOG	Read	40051
Run Hours- High*	Hrs.	INTEGER	Read	45089
Run Hours- Low*	Hrs.	INTEGER	Read	45088
Number Of Starts H*	-	INTEGER	Read	45094
Number Of Starts L*	-	INTEGER	Read	45095
Compressor Status	-	Boolean	Read	10018
Run Hours Total	-	INTEGER	Read	45391

Protocol Details	
Parity	None
Baud rate	19200
Protocol	Modbus/Modbus Extended
Start bit	1
Stop bit	1
No of bit	8
Communication Interface	RS-485
Interface Connection	2 Wire

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The connection with the RS485 network is carried out using the plug-in terminal connector on the card. Pin-wiring of the connector is stamped on the card. If the card is placed in the last position of the supervision serial line, pins 2 and 3, you must connect a 120 Ω - 1/4 W end line resistor, as shown.



pin	significato <i>meaning</i>
1	GND
2	RX+/TX+
3	RX-/TX-

*Total Run hours/No of starts will be calculated as below *Total run hrs = Run hour H (45023) x 1000 + Run hour L (45024) *Total no of starts = Number of starts H (45019) x 1000 + Number of starts L (45020)